Department of Computing

Course Code: **CS332** — Class: **BSCS-13**

Lab 08

Getting Started with Apache Solr

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Introduction

Solr is the popular, blazing-fast, open-source **NoSQL search platform** from the Apache Lucene project. Its major features include powerful full-text search, hit highlighting, faceted search, dynamic clustering, database integration, rich document handling, and geospatial search. Solr is highly scalable, providing fault-tolerant distributed search and indexing, and powers the search and navigation features of many of the world's largest internet sites.

Objectives

To install, set up, and understand the basics of Apache Solr including standalone mode and distributed SolrCloud cluster creation with shards and replicas.

Tools / Software Requirement

- Ubuntu 24.04.1 LTS (Noble)
- OpenJDK 21.0.10
- Apache Solr 8.11.4

Description

Solr is an open-source search platform built on top of **Apache Lucene**. It was created by Yonik Seely in 2004 and became an Apache open-source project in January 2006. Solr can be used alongside Hadoop for large-scale data processing. It is not only a search engine but also a non-relational NoSQL data store.

Key Features of Apache Solr

1. **RESTful APIs** – Communicate via XML, JSON, or CSV without needing Java skills.
2. **Full-Text Search** – Tokens, phrases, spell check, wildcard, and auto-complete.
3. **Enterprise Ready** – Deployable as standalone, distributed, or cloud.
4. **Flexible & Extensible** – Customize components by extending Java classes.
5. **NoSQL Database** – Distribute search tasks across a cluster like a big-data NoSQL DB.
6. **Admin Interface** – Feature-rich UI for managing logs, documents, and queries.
7. **Highly Scalable** – Scale by adding replicas when used with Hadoop.
8. **Relevance Sorting** – Results delivered in order of relevance to the user's query.

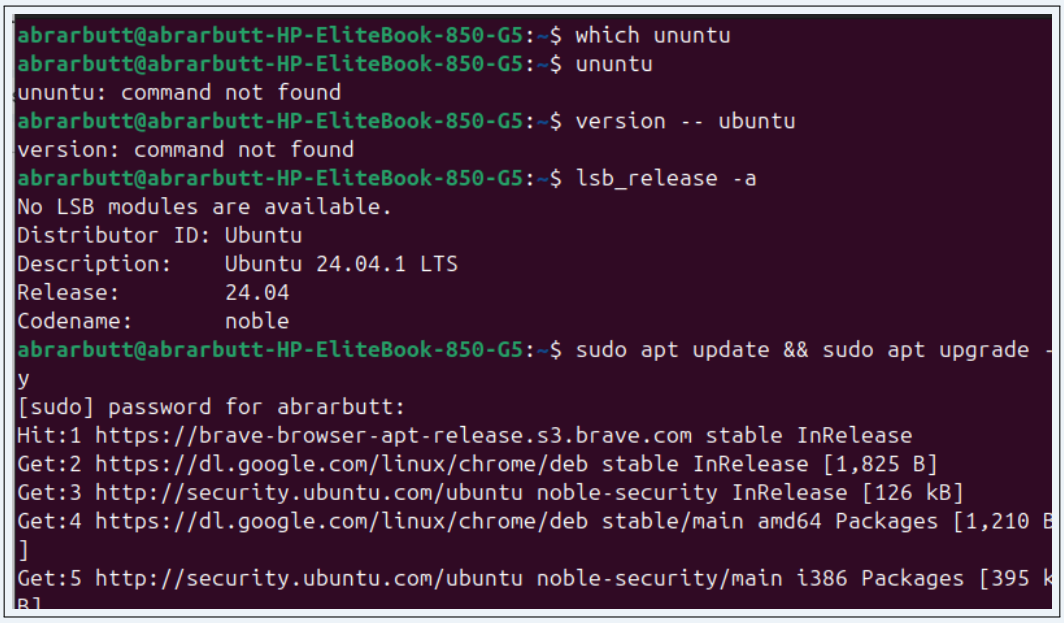
Lab Tasks

Task 1 – Install & Configure Apache Solr on Ubuntu

Step 1: Check Ubuntu Version & Update System

First, the Ubuntu version was confirmed using `lsb_release -a`, showing **Ubuntu 24.04.1 LTS**. The system was then updated with:

```
lsb_release -a
sudo apt update && sudo apt upgrade -y
```



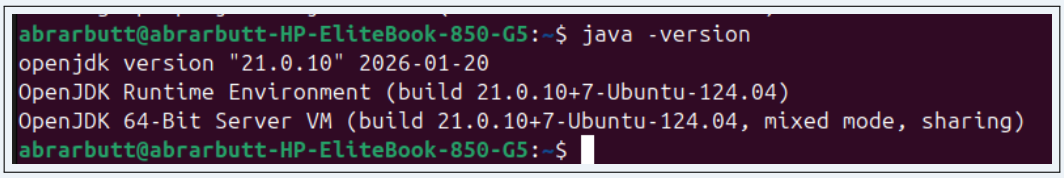
Ubuntu 24.04.1 LTS confirmed and system update initiated

Step 2: Install Java (OpenJDK 21)

Apache Solr requires Java. OpenJDK was installed and verified:

```
sudo apt install openjdk-11-jdk -y
java -version
```

Output confirmed: `openjdk version "21.0.10" 2026-01-20`



Java version confirmed – OpenJDK 21.0.10 running on Ubuntu

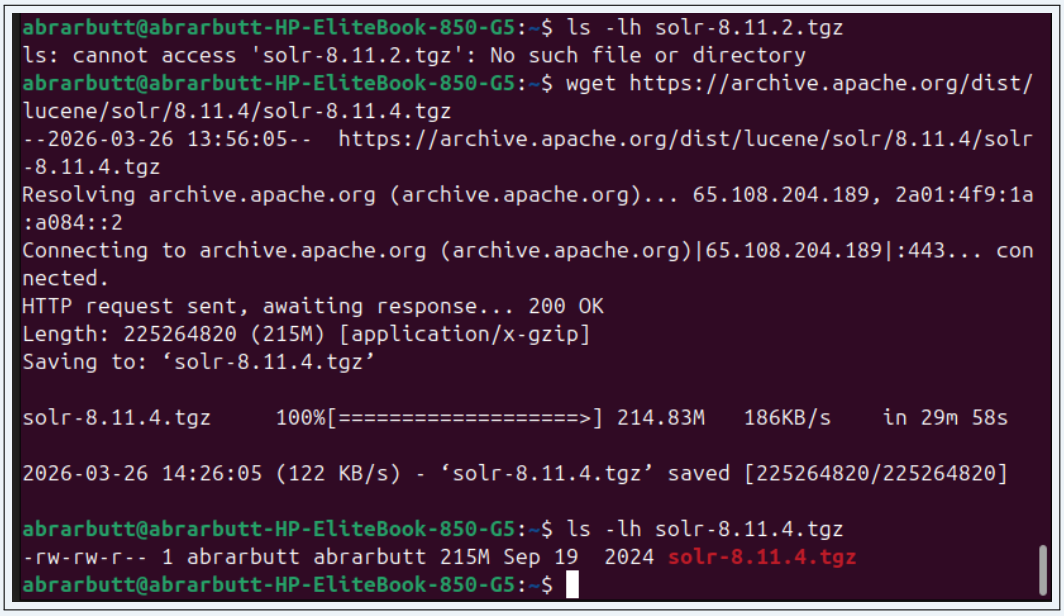
Task 1 – Continued

Step 3: Download Apache Solr 8.11.4

The initial URL for Solr 8.11.2 returned a 404 error. The correct file was fetched from the Apache archive:

```
wget https://archive.apache.org/dist/lucene/solr/8.11.4/solr
    -8.11.4.tgz
ls -lh solr-8.11.4.tgz
```

The 215 MB file was downloaded successfully.



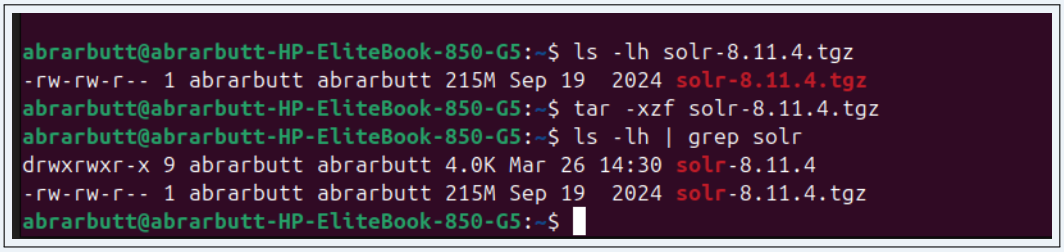
```
abrarbutt@abrarbutt-HP-EliteBook-850-G5:~$ ls -lh solr-8.11.2.tgz
ls: cannot access 'solr-8.11.2.tgz': No such file or directory
abrarbutt@abrarbutt-HP-EliteBook-850-G5:~$ wget https://archive.apache.org/dist/
lucene/solr/8.11.4/solr-8.11.4.tgz
--2026-03-26 13:56:05--  https://archive.apache.org/dist/lucene/solr/8.11.4/solr
-8.11.4.tgz
Resolving archive.apache.org (archive.apache.org)... 65.108.204.189, 2a01:4f9:1a
:a084::2
Connecting to archive.apache.org (archive.apache.org)|65.108.204.189|:443... con
nected.
HTTP request sent, awaiting response... 200 OK
Length: 225264820 (215M) [application/x-gzip]
Saving to: ‘solr-8.11.4.tgz’

solr-8.11.4.tgz     100%[===================>] 214.83M   186KB/s    in 29m 58s

2026-03-26 14:26:05 (122 KB/s) - ‘solr-8.11.4.tgz’ saved [225264820/225264820]

abrarbutt@abrarbutt-HP-EliteBook-850-G5:~$ ls -lh solr-8.11.4.tgz
-rw-rw-r-- 1 abrarbutt abrarbutt 215M Sep 19  2024 solr-8.11.4.tgz
abrarbutt@abrarbutt-HP-EliteBook-850-G5:~$
```

Solr 8.11.4 downloaded successfully (215M) from Apache archive

Step 4: Extract Solr Archive

```
tar -xzf solr-8.11.4.tgz
ls -lh | grep solr
```

The extracted folder `solr-8.11.4` appeared alongside the archive file.



```
abrarbutt@abrarbutt-HP-EliteBook-850-G5:~$ ls -lh solr-8.11.4.tgz
-rw-rw-r-- 1 abrarbutt abrarbutt 215M Sep 19  2024 solr-8.11.4.tgz
abrarbutt@abrarbutt-HP-EliteBook-850-G5:~$ tar -xzf solr-8.11.4.tgz
abrarbutt@abrarbutt-HP-EliteBook-850-G5:~$ ls -lh | grep solr
drwxrwxr-x 9 abrarbutt abrarbutt 4.0K Mar 26 14:30 solr-8.11.4
-rw-rw-r-- 1 abrarbutt abrarbutt 215M Sep 19  2024 solr-8.11.4.tgz
abrarbutt@abrarbutt-HP-EliteBook-850-G5:~$
```

Solr archive extracted – solr-8.11.4 directory created (4.0K)

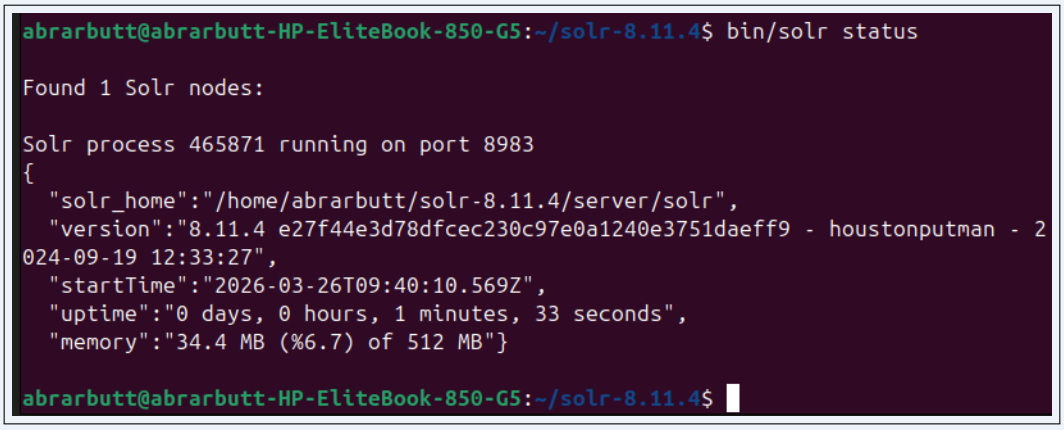
Task 1 – Continued

Step 5: Start Apache Solr & Verify Status

```
cd solr-8.11.4
bin/solr start
bin/solr status
```

Solr started on port **8983**. Status showed:

- Process ID: 465871
- Version: 8.11.4
- Memory: 34.4 MB of 512 MB
- Uptime: 1 minute, 33 seconds

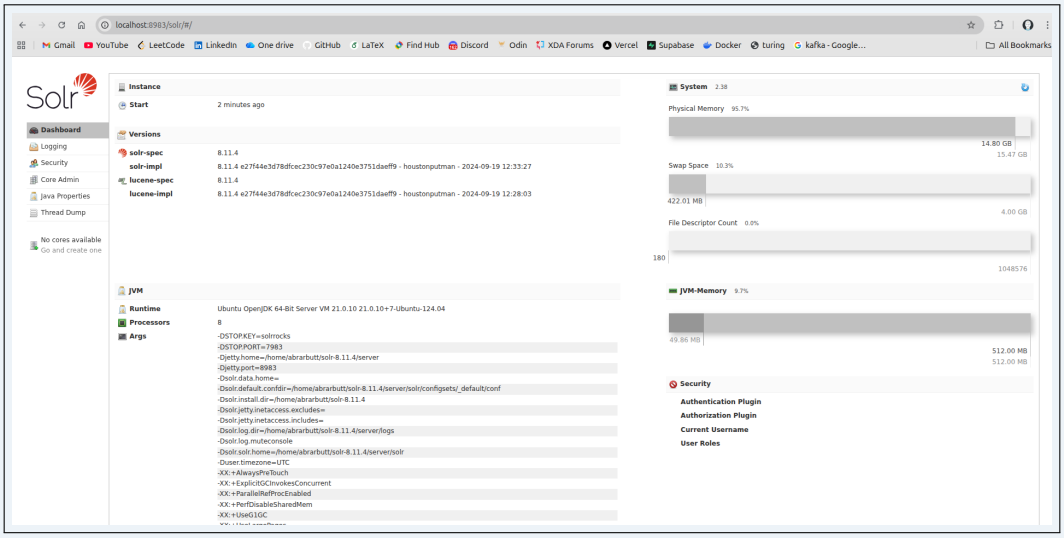


```
abrarbutt@abrarbutt-HP-EliteBook-850-G5:~/solr-8.11.4$ bin/solr status

Found 1 Solr nodes:

Solr process 465871 running on port 8983
{
  "solr_home":"/home/abrarbutt/solr-8.11.4/server/solr",
  "version":"8.11.4 e27f44e3d78dfcec230c97e0a1240e3751daeff9 - houstonputman - 2
024-09-19 12:33:27",
  "startTime":"2026-03-26T09:40:10.569Z",
  "uptime":"0 days, 0 hours, 1 minutes, 33 seconds",
  "memory":"34.4 MB (%6.7) of 512 MB"}

abrarbutt@abrarbutt-HP-EliteBook-850-G5:~/solr-8.11.4$
```

Solr running on port 8983 – status confirmed with version and memory info

Step 6: Open Solr Admin UI in Browser

Navigated to `http://localhost:8983/solr` in the browser.

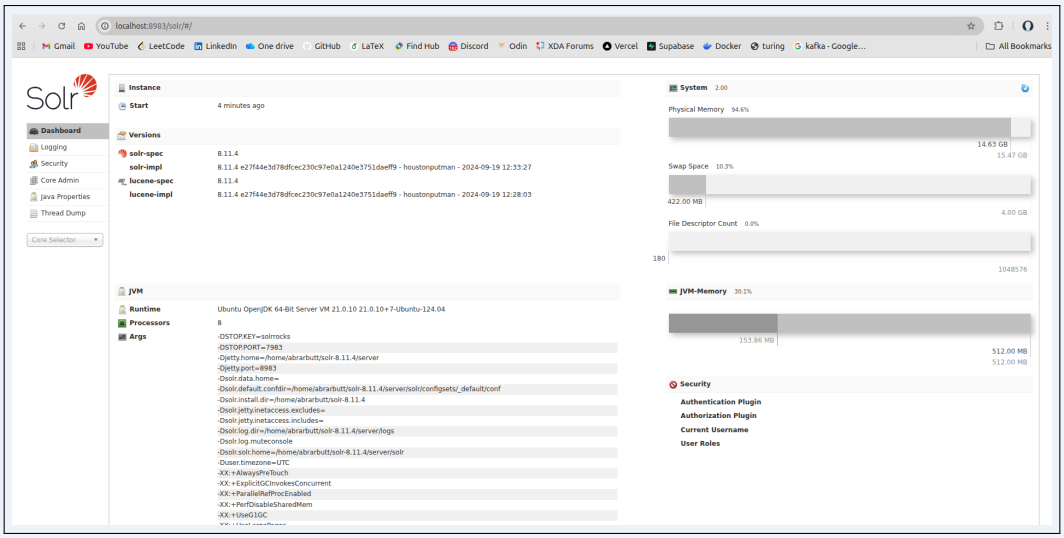


Apache Solr Admin Dashboard loaded at localhost:8983 – showing version 8.11.4, JVM info, and system memory

Task 1 – Continued

Step 7: Admin UI Verified (Core Selector Available)

The Admin UI was confirmed working. The left panel showed navigation options including Logging, Security, Core Admin, Java Properties, and Thread Dump.



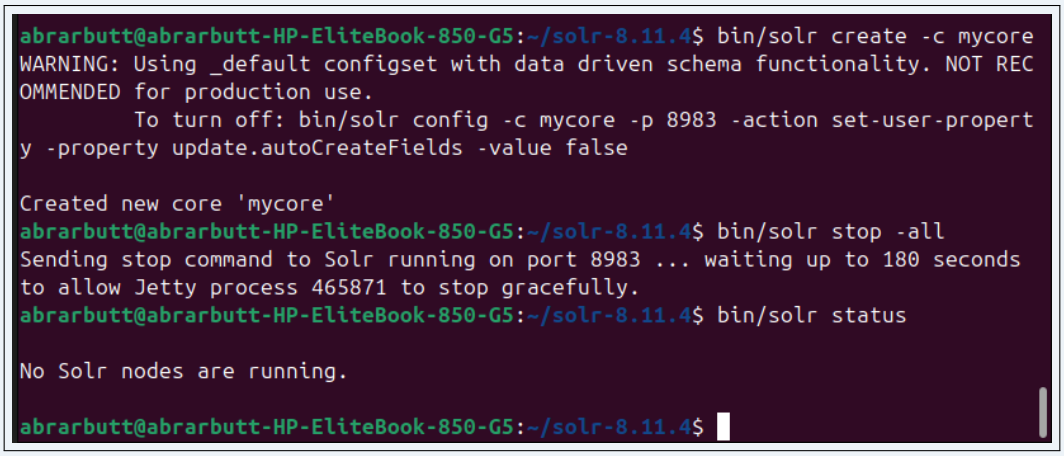
Solr Admin UI – Core Selector visible in left panel, system running normally

Step 8: Create Core & Stop Solr

A new core named `mycore` was created, then Solr was stopped cleanly:

```
bin/solr create -c mycore
bin/solr stop -all
bin/solr status
```

Output confirmed: *"Created new core 'mycore"'* followed by *"No Solr nodes are running."*



```
abrarbutt@abrarbutt-HP-EliteBook-850-G5:~/solr-8.11.4$ bin/solr create -c mycore
WARNING: Using _default configset with data driven schema functionality. NOT REC
OMMENDED for production use.
         To turn off: bin/solr config -c mycore -p 8983 -action set-user-propert
y -property update.autoCreateFields -value false

Created new core 'mycore'
abrarbutt@abrarbutt-HP-EliteBook-850-G5:~/solr-8.11.4$ bin/solr stop -all
Sending stop command to Solr running on port 8983 ... waiting up to 180 seconds
to allow Jetty process 465871 to stop gracefully.
abrarbutt@abrarbutt-HP-EliteBook-850-G5:~/solr-8.11.4$ bin/solr status

No Solr nodes are running.

abrarbutt@abrarbutt-HP-EliteBook-850-G5:~/solr-8.11.4$
```

mycore created successfully – Solr stopped cleanly, no nodes running

Task 1 Complete – Apache Solr installed, configured, Admin UI verified, and core created.

Task 2 – Create a Cluster of Servers (SolrCloud with Shards)

Overview: SolrCloud is the distributed, fault-tolerant mode of Solr. It uses Apache ZooKeeper for cluster coordination. The cluster consists of:

- **Collection:** A complete logical index (`mycluster`)
- **Shards:** Logical partitions of the collection (2 shards)
- **Replicas:** Physical copies of each shard (2 replicas per shard)
- **Nodes:** Node 1 on port 8983, Node 2 on port 7574
- **ZooKeeper:** Embedded on port 9983

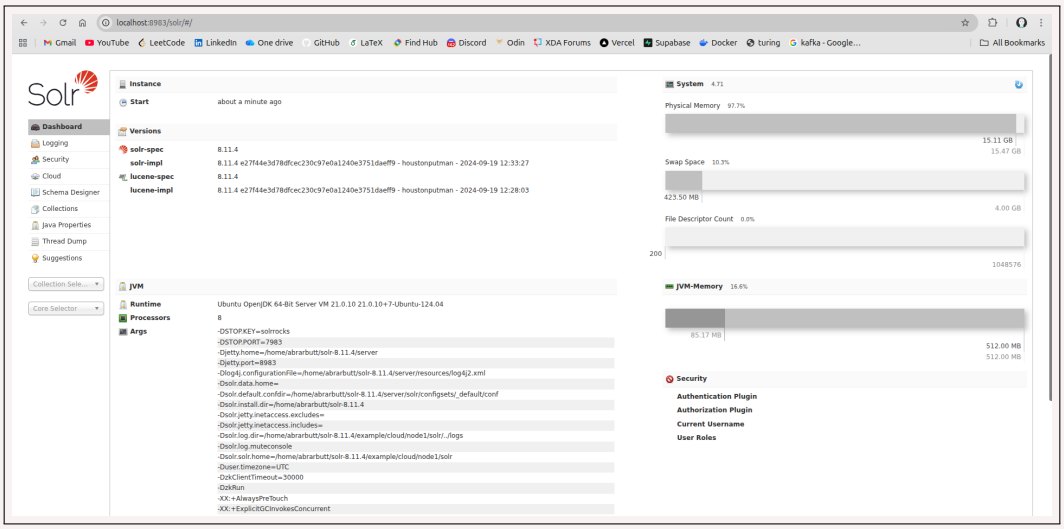
Step 1: Start SolrCloud Interactive Wizard

```
bin/solr start -e cloud
```

Wizard answers: 2 nodes, ports 8983 & 7574, collection name: `mycluster`, 2 shards, 2 replicas, default config.

Step 2: Verify SolrCloud Admin UI (Node 1 – Port 8983)

After the wizard completed, the Admin UI at `http://localhost:8983/solr` now showed **Cloud**, Collections, and Schema Designer in the left panel – confirming SolrCloud mode.



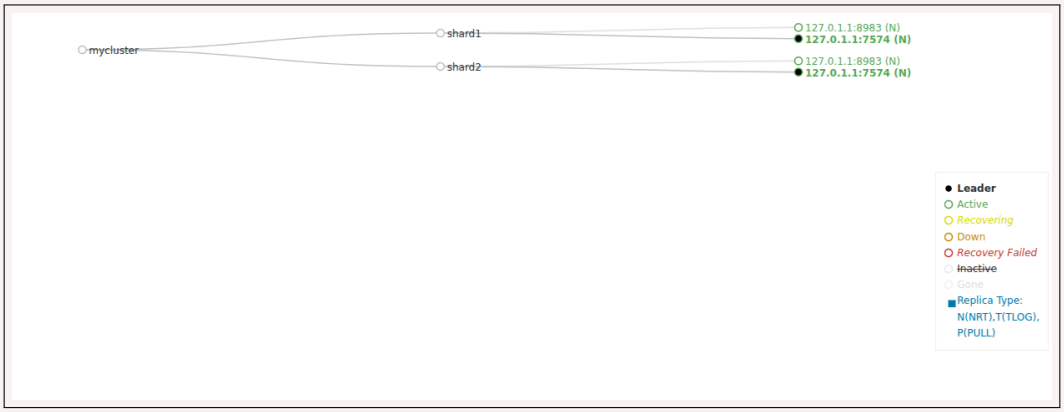
SolrCloud Admin UI at port 8983 – Cloud menu visible, mycluster collection created

Task 2 – Continued

Step 3: View Cluster Graph (Shards & Replicas)

Navigated to **Cloud** → **Graph** in the Admin UI. The graph shows:

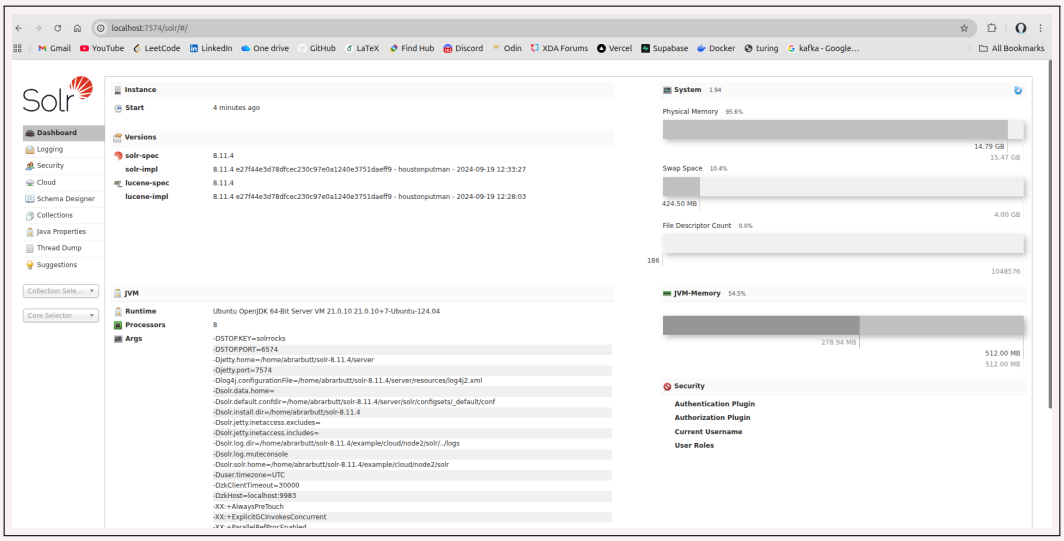
- Collection `mycluster` split into **shard1** and **shard2**
- Each shard has 2 replicas: on `127.0.1.1:8983` and `127.0.1.1:7574`
- Leader replicas shown with filled circle (•)
- Active replicas shown with green hollow circle



SolrCloud Graph – mycluster with shard1 and shard2, each having replicas on both nodes (8983 & 7574)

Step 4: Verify Node 2 is Running (Port 7574)

Opened `http://localhost:7574/solr` in the browser to confirm the second node is active.



Node 2 Admin UI at localhost:7574 – Solr 8.11.4 running, Cloud menu visible, 512 MB JVM, 8 processors

Task 2 – Continued

Step 5: Index Documents into the Cluster

Sample book documents were indexed into the `mycluster` collection:

```
bin/post -c mycluster example/exampledocs/books.json
```

Output confirmed:

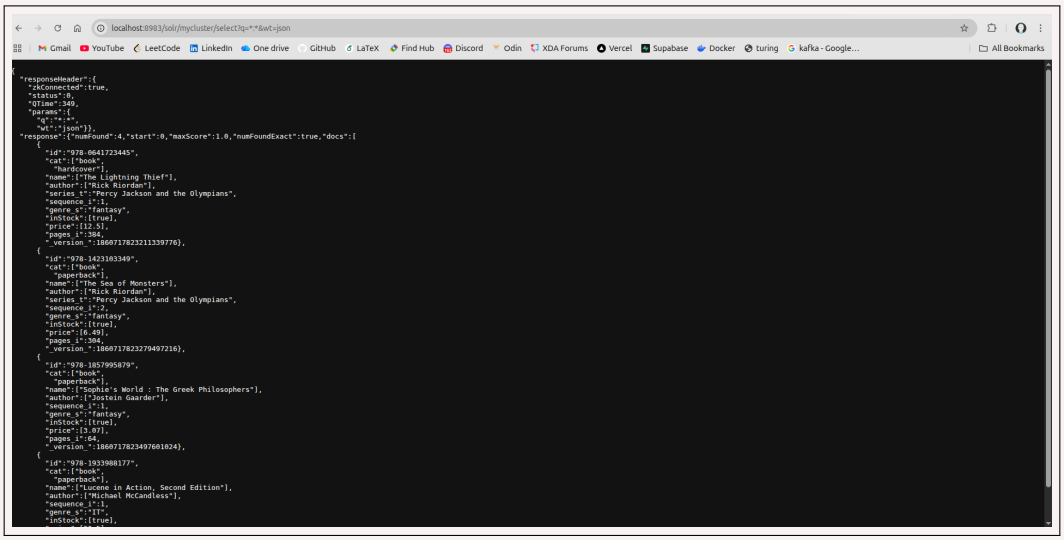
- `1 files indexed.`
- `COMMITting Solr index changes...`
- `Time spent:  0:00:07.130`

Step 6: Query the Distributed Collection

Navigated to the following URL to verify documents are searchable across shards:

```
http://localhost:8983/solr/mycluster/select?q=*:*&wt=json
```

Result: `"numFound":4` – 4 books returned (The Lightning Thief, The Sea of Monsters, Sophie's World, Lucene in Action).



JSON query results from mycluster – numFound:4, documents distributed across both shards

Task 2 – Continued

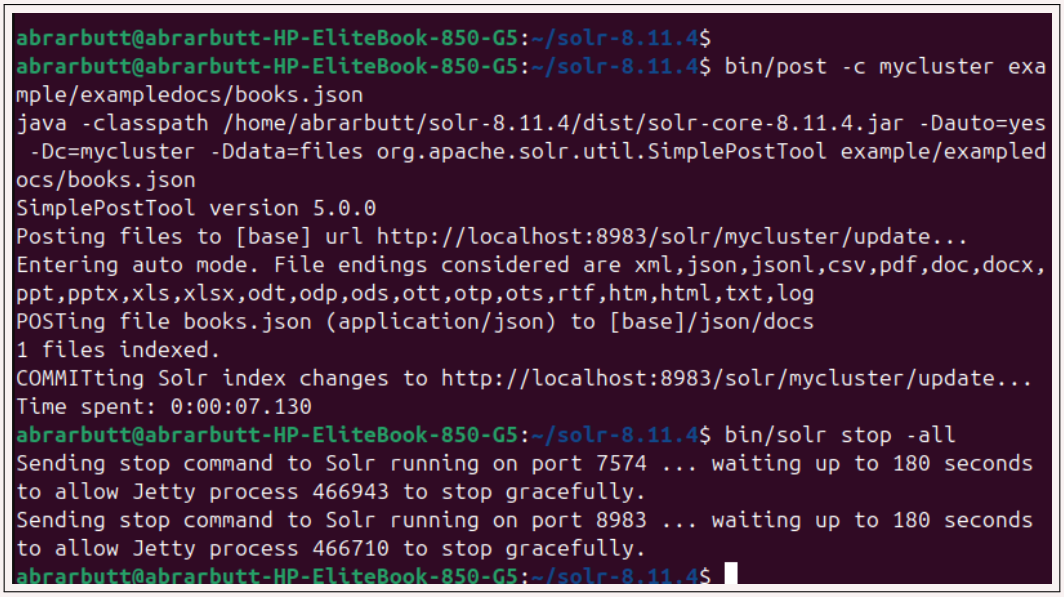
Step 7: Index Documents & Stop Cluster

Final step – documents indexed and both nodes stopped cleanly:

```
bin/post -c mycluster example/exampledocs/books.json
bin/solr stop -all
```

Both nodes stopped gracefully:

- Port 7574 – Jetty process 466943 stopped
- Port 8983 – Jetty process 466710 stopped



```
abrarbutt@abrarbutt-HP-EliteBook-850-G5:~/solr-8.11.4$
abrarbutt@abrarbutt-HP-EliteBook-850-G5:~/solr-8.11.4$ bin/post -c mycluster exa
mple/exampledocs/books.json
java -classpath /home/abrarbutt/solr-8.11.4/dist/solr-core-8.11.4.jar -Dauto=yes
 -Dc=mycluster -Ddata=files org.apache.solr.util.SimplePostTool example/exampled
ocs/books.json
SimplePostTool version 5.0.0
Posting files to [base] url http://localhost:8983/solr/mycluster/update...
Entering auto mode. File endings considered are xml,json,jsonl,csv,pdf,doc,docx,
ppt,pptx,xls,xlsx,odt,odp,ods,ott,otp,ots,rtf,htm,html,txt,log
POSTing file books.json (application/json) to [base]/json/docs
1 files indexed.
COMMITting Solr index changes to http://localhost:8983/solr/mycluster/update...
Time spent: 0:00:07.130
abrarbutt@abrarbutt-HP-EliteBook-850-G5:~/solr-8.11.4$ bin/solr stop -all
Sending stop command to Solr running on port 7574 ... waiting up to 180 seconds
to allow Jetty process 466943 to stop gracefully.
Sending stop command to Solr running on port 8983 ... waiting up to 180 seconds
to allow Jetty process 466710 to stop gracefully.
abrarbutt@abrarbutt-HP-EliteBook-850-G5:~/solr-8.11.4$
```

Documents indexed (1 file, 7.1s), then both SolrCloud nodes stopped cleanly on ports 7574 and 8983

Task 2 Complete – SolrCloud cluster with 2 nodes, 2 shards, 2 replicas created, documents indexed & queried successfully.

Summary

Item	Details
OS	Ubuntu 24.04.1 LTS (Noble)
Java Version	OpenJDK 21.0.10
Solr Version	Apache Solr 8.11.4
Task 1	Installed Solr, started standalone server on port 8983, opened Admin UI, created `mycore`
Task 2	Started SolrCloud with 2 nodes (8983, 7574), ZooKeeper on 9983, collection `mycluster` with 2 shards × 2 replicas, indexed 4 documents, queried across cluster
Query Result	numFound: 4 (The Lightning Thief, Sea of Monsters, Sophie's World, Lucene in Action)
Final State	All nodes stopped cleanly